Dice Probability

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1725 | Mathematics | https://cses.fi/problemset/task/1725

You throw a dice n times, and every throw produces an outcome between 1 and 6. What is the probability that the sum of outcomes is between a and b?

Input

The only input line contains three integers n, a and b.

Output

Print the probability rounded to six decimal places (rounding half to even).

Constraints

- $1 \le n \le 100$
- $1 \le a \le b \le 6n$

Example

Input
`2 9 10`

Output
`0.194444`